QRAFT

Quantum-Resistant
Anonymous Frame Transactions

Private ETH payments from an ERC-8286 PQ wallet.

Local proofs

Encrypted receiving

EIP-8141 transactions

Two balances. Different visibility.

Public smart-account balance

ETH held by your ERC-8286 PQ account.
The address and balance are public.

Faucet funding arrives here.

Private pool balance

Encrypted notes belonging to you.
Your device calculates the available total.

Deposit to enter. Withdraw to a public address.

Separate notes.
One private balance.

The ERC-8286 PQ wallet finds your notes, tracks spends and selects payment inputs.

0.00002 ETH
Deposit one

0.00002 ETH
Deposit two

AVAILABLE PRIVATELY

0.00004 ETH

+ Add ETH ↗ Send ↙ Receive

Illustrated from the verified demo flow.

One passkey unlocks the ERC-8286 PQ wallet.

ML-DSA-44

Signs the public account’s EIP-8141 transactions.

ML-KEM-768

Encrypts incoming notes for the receiving key.

Spending secret

Proves note ownership and derives nullifiers.

Derived from one master secret. The encrypted backup still requires the original passkey.

Add ETH to the private balance.

Open the ERC-8286 PQ wallet dialog.

Check the public address and ETH balance.

Deposit 0.00002 ETH twice.

Gas is sponsored. The deposit address and amount remain public.

TWO CONFIRMED DEPOSITS

Deposit 1	0.00002 ETH
Deposit 2	0.00002 ETH

Available privately

0.00004 ETH

Send an exact amount. Keep the change private.

Alice’s two notes

0.00004 ETH

Selected automatically.

→

Bob receives	**0.00003 ETH**
Alice’s change	**0.00001 ETH**

Two new encrypted notes replace the inputs.

Bob shares a qraft1.... receiving address. His ERC-8286 PQ account need not be deployed to receive.

Receive privately.
Withdraw to a public address.

Bob’s private balance

0.00003

ETH discovered and decrypted locally.

Public withdrawal

0.00001

ETH sent to the chosen 0x address.

Private remainder

0.00002

ETH retained as a change note.

The withdrawal address and amount are public. The private remainder stays encrypted.

Three frames.
One EIP-8141 transaction.

01 · VERIFY

PQ authorization

Check the submitting ERC-8286 account’s ML-DSA signature.

02 · VERIFY

Gas sponsorship

Validate the permitted transaction shape and gas limits.

03 · SENDER

Pool settlement

Verify the proof, consume nullifiers and create outputs or pay out.

Private sends use a shared carrier. Deposits use the user’s account. The external faucet is a type-2 exception.

A tested prototype.
A defined security boundary.

Verified

90 tests and a live UI-to-chain flow.

Replay rejection, failed-payout rollback, lost-response recovery and exact balances.

Current limits

Groth16 proofs use conventional cryptography and a development setup.

Nym covers reads and deposits. The HTTPS relay can observe IP and timing.

Live E2E uses direct test reads. Full uninterrupted Nym E2E is not yet verified. No production-security claim.

Private ETH payments. ERC-8286 PQ accounts. EIP-8141 execution.

Unified private balance. Local proofs. Encrypted receiving.

`github.com/Shoydon/QRAFT`

Next: anonymize relay submission and complete independent security review.